PROFETA Analysis Report

Universal Multi-Domain Hybrid Prediction System

Version 5.0.0 | Enterprise Edition

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PREDICTIONS	CONFIDENCE	TREND	R^2 SCORE
13340	60%	UP	0.48

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Enterprise AI Solutions

1. Executive Summary

This report presents the analysis results from **PROFETA Universal v5.0**, an enterprise-grade time series prediction system utilizing a multi-head LSTM ensemble architecture with **20 models**.

Key Findings:
The system generated **13340 predictions** over a **13340.0-hour** time horizon. The dominant predicted trend is **bullish** with an expected price change of **+0.17%**. The overall prediction confidence is **moderate** at **60.3%**.

Model Performance:
The regression model achieved an R^2 score of **0.4790** with RMSE of **226.591066**. The classification module achieved **38.7%** accuracy with F1-score of **0.2140**. The agreement rate between regression and classification heads is **7.2%**.

1.1 Configuration Overview

Parameter	Value
Configuration File	N/A
Domain Type	financial / crypto
Input Granularity	auto
Output Granularity	hour
Classification	ENABLED
Ensemble Models	20
Ensemble Delta	-0.122212

2. Performance Metrics

2.1 Comprehensive Metrics

Metric Category	Metric	Value	Status
REGRESSION	RMSE	226.591066	—
	MAE	188.317136	—
	MAPE	0.1658%	—
	R^2	0.4790	✗ Needs Improvement
	Mean Delta	-0.000066	—
	Std Delta	226.591066	—
CLASSIFICATION	Accuracy	38.72%	✗ Needs Improvement
	Precision (Macro)	0.2622	—
	Recall (Macro)	0.3402	—
	F1-Score (Macro)	0.2140	—

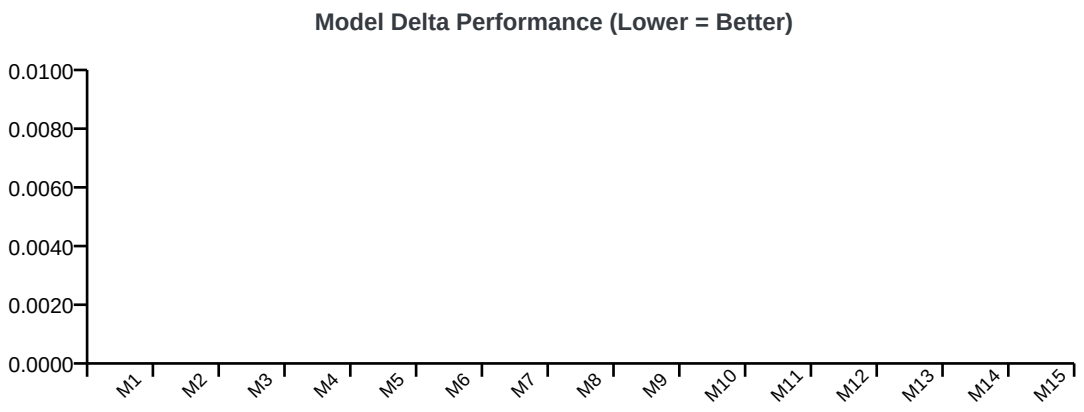
3. Ensemble Model Analysis

The PROFETA ensemble consists of **20 LSTM models** with diverse architectures. Each model contributes to the final prediction through weighted averaging. The ensemble delta (standard deviation across model predictions) is **-0.122212**, indicating low prediction variance.

3.1 Top Performing Models

Rank	Model	Seq Len	LSTM Units	Dropout	BiDir	Attn	Delta	Epochs
#1	MODEL_1	60	64	20.0%	✓	✗	0.00000	0
#2	MODEL_2	60	64	20.0%	✗	✗	0.00000	0
#3	MODEL_3	60	128	40.0%	✓	✗	0.00000	0
#4	MODEL_4	60	128	40.0%	✗	✗	0.00000	0
#5	MODEL_5	60	32	30.0%	✓	✗	0.00000	0
#6	MODEL_6	60	32	30.0%	✗	✗	0.00000	0
#7	MODEL_7	60	128	30.0%	✓	✗	0.00000	0
#8	MODEL_8	60	32	10.0%	✗	✗	0.00000	0
#9	MODEL_9	60	32	30.0%	✓	✗	0.00000	0
#10	MODEL_10	60	512	20.0%	✗	✗	0.00000	0

3.2 Model Delta Comparison



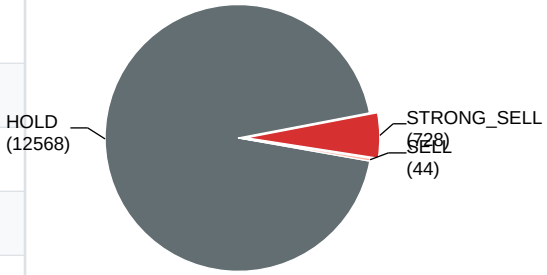
4. Prediction Analysis

4.1 Prediction Summary

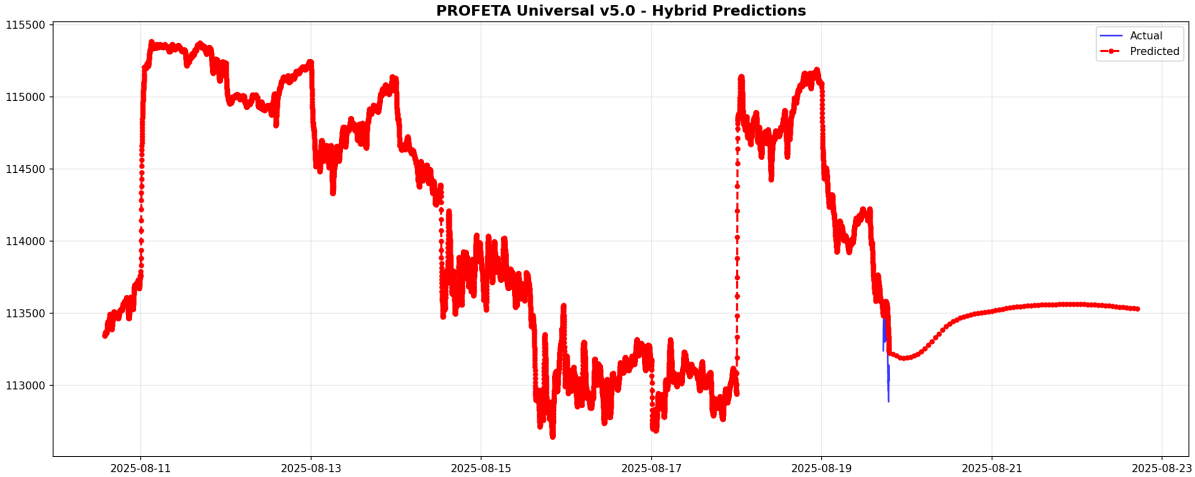
Parameter	Value
Total Predictions	13340
Time Horizon	13340.0 hours
First Prediction	2025-08-10 13:51:00+00:00
Last Prediction	2025-08-22 17:00:00
Starting Price	113,343.28
Ending Price (Predicted)	113,530.87
Expected Change	**+0.17%**
Dominant Trend	**UP**
Average Confidence	60.3%
Reg/Cls Agreement Rate	7.2%

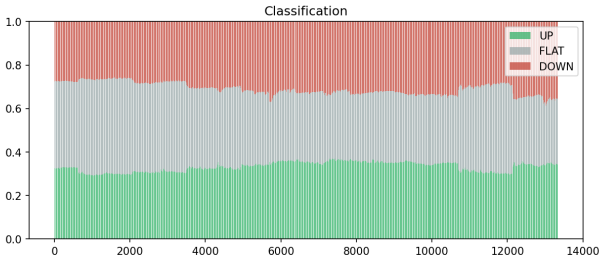
4.2 Trading Signal Distribution

Signal	Count	Percentage	Indicator
STRONG_BUY	0	0.0%	■■
BUY	0	0.0%	■
HOLD	12568	94.2%	■
SELL	44	0.3%	■
STRONG_SELL	728	5.5%	■■



4.3 Prediction Visualization





REGRESSION

RMSE: 226.591066
MAE: 188.317136
R^2: 0.4790

CLASSIFICATION

Accuracy: 38.7%
F1: 0.2140

5. Generated Output Files

File Type	Filename	Description
CSV	real_time_ens_hours.csv	Raw predictions with all columns
JSON	predictions.json	Structured data with metadata and metrics
PNG	analysis.png	Visualization chart
PDF	This report	Comprehensive analysis report

5.1 Execution Timing

Phase	Duration
Training	0.00 seconds
Prediction	0.00 seconds
Total Execution	0.00 seconds

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